

DEPARTMENT OF MINING ENGINEERING
IIT (ISM), Dhanbad

LECTURE PLAN

Course: IV B. Tech (Mining Engineering)

Session: 2020-21, **Semester:** Winter

Subject: Mine Ventilation (MNC 206)

L-T-P : 3-1-0

Subject Teacher: Prof. D. P. Mishra

Unit No.	Topics	Lecture & Tutorial Hours
1	Composition of mine atmosphere: Mine gases - production, properties and physiological effects; Sampling and analysis of mine air; Methane content of coal seams; Methane emission; Methane layering; Methane drainage; Detection of methane - Flame safety lamp, Methanometer; Radon gas and its daughter products; Monitoring of mine gases.	11
2	Heat and humidity: Sources of heat in mines; Effects of heat and humidity; Psychrometry, Air cooling power, Kata thermometer; Air-conditioning.	7
3	Air flow through mine openings: Laws of fluid flow, resistance of mine airways, equivalent orifice, losses in airways, splitting of air current, economic design of airways; Ventilation flow control devices; Permissible air velocities in different types of workings/openings; Standards of ventilation.	10
4	Natural ventilation: Causes, effect of seasonal variations, calculation of NVP from air densities, thermodynamic principles and other methods.	6
5	Mechanical ventilation: Types of mine fans; Theory, characteristics and suitability of fans; Selection and output control of fan; Fans in series and parallel; Forcing and exhaust configurations; Reversal of airflow; Fan drift, Diffuser, Evasee; Booster fan and auxiliary ventilation; Venturi blowers; Ventilation of deep mines, underground and open pit.	10
6	Ventilation planning: Planning of ventilation systems and economic considerations; Ventilation layouts for underground coal and metal mines; Calculation of air quantity required for ventilating a mine; Calculation of total mine head; Ventilation network analysis principles and computer applications; Ventilation surveys.	8
Total		52


(D. P. Mishra)
Assoc. Professor

Text books:

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| 1. Mine Environment and Ventilation | : Prof. G. B. Misra |
| 2. Mine Ventilation | : Prof. S. P. Banerjee |
| 3. Subsurface Ventilation and Environmental Engineering | : Prof. M. J. McPherson |
| 4. Environmental Engineering in Mines | : Dr. V. S. Vutkuri and Dr. R. D. Lama |
| 5. Mine Ventilation and Air Conditioning | : Prof. H. L. Hartman, Prof. Jan Mutmansky and Prof. Y. J. Wang |
| 6. Mine Environmental Engineering, Vol. 1 & Vol. 2 | : Prof. Mritunjoy Sengupta |